

Porous materials as open volumetric solar receivers: Experimental determination of thermophysical and heat transfer properties

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Abstract

Porous solids like extruded monoliths with parallel channels and thin walls made from various oxide and non-oxide ceramics, ceramic foams and metal structures have been tested in the past with the objective of applying them as open volumetric receivers in concentrated solar radiation. In this application, ambient air flows through the solid, which is heated by concentrated solar radiation. A heat exchanger then transfers the energy to a conventional steam turbine process. In all cases, to obtain high efficiencies, high absorptivity in the visible and near infrared range has to be combined with a high porosity to create large surfaces for convective heat transfer from the solid absorber to the fluid. However, it can be shown that especially high performance absorbers tend to be sensitive to inhomogeneous flux distributions, which may cause local overheating of the material. In various tests with specific kinds of materials, flow instabilities occurred, which partly leads to hot spots and a sudden destruction of the receiver. To achieve both high efficiencies and reliable operation, an optimised combination of geometrical as well as thermal conductivity and heat transfer parameters has to be selected. A precise knowledge of these quantities for a number of various materials is necessary to estimate the limits for stable flow conditions on the basis of complex numerical simulation programs. Finally, efficiency and performance tests with candidate materials have been carried out. In this paper, the experimental work on a variety of porous materials is reported. The paper will report on methodology and results of thermal conductivity, convective heat transfer coefficient and efficiency measurements of these monolithic materials. It will also present an experimental set-up designed to investigate how the properties of the porous materials affect flow stability. Based on these results, a recommendation for the design of volumetric absorbers will be given.

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Nomenclature

Λ	thermal conductivity (W/mK)
κ	thermal diffusivity (m ² /s)
C	specific heat (W/kg K)
α	heat transfer coefficient (W/m ² K)
A_V	specific surface (m ² /m ³)
ρ	Density (kg/m ³)
v	air flow velocity (m/s)
v_{Pore}	pore size level air flow velocity (m/s)
Re	Reynolds number
P_0	porosity (%)

1. Introduction

Research and development on volumetric receivers has been carried out for more than two decades. However, solar thermal concentrating technology is still far away from economic competitiveness. Only an artificial market can lead to a market introduction. Such an artificial market is presently being created by the Spanish government with their proposal to pay about 16 cents/kWh for solar electricity. Consequently, a European consortium is planning a 10 MW solar installation in southern Spain called Sanlucar Solar [1]. It makes use of the so-called solar tower technology, which has been successfully tested in several projects during the last 20 years, among which the famous American *Solar One* and *Solar Two* [2], the Spanish *CESA 1* and the French *Themis* projects have to be mentioned [3].

The central element of the most common tower technology is the open volumetric receiver, consisting of a porous material heated by concentrated solar radiation. Ambient air flowing through the porous material is heated to temperatures around 700 °C and is used to feed a conventional steam turbine process. Many studies have been conducted on properties and physical restrictions of the open volumetric receiver principle [4,5] and past receiver concepts have been based on materials like metallic wire meshes, catalyst carriers with parallel channels for exhaust systems, or ceramic foam materials, which have been optimised for other applications.

In this paper, new approaches are reported, which have been started in the two projects SOLPOR and PORENKÖRPER, in which energy technology experts have co-operated with material scientists and fluid mechanic experts to create novel volumetric receiver designs. Some of these approaches have already been realised at the laboratory scale and tested in concentrated radiation.

Secondly, the influence of heat transfer and thermal conductivity properties on the performance of volumetric receivers in practical applications has been investigated. For this purpose, these quantities have been measured for a couple of materials.

2. Experimental

2.1. Efficiency

Efficiency tests have been carried out in concentrated radiation using the DLR “Solar Furnace”, an installation consisting of a 25 m² heliostat and a fixed concentrator [6]. In the focus of the furnace, an isolated test bed is used, in which absorber samples can be placed (Fig. 1). A fan forces ambient air to flow through the sample followed by a water heat exchanger. The power being transferred to the water circuit and the power remaining in the air can be calculated from temperature measurements.

2.2. Heat transfer coefficient

Heat transfer measurements have been undertaken in a test bed, in which an air flow of alternating temperature flows through the porous sample (Fig. 2). The air temperature can be seen as a temperature wave $T(t, x)$. The sample induces a phase shift $\Delta\phi$ and an attenuation $F(\omega) = \max(T(L, t)) / \max(T(0, t))$ of the temperature wave. From these, the product of the specific surface A_V and the heat transfer coefficient α can be calculated [7].

2.3. Flow instability

Flow instability is primarily investigated in a non-stationary experiment sketched in Fig. 3. Samples of cylindrical shape are heated up to 800 °C in a tube heater at zero fluid velocity v . At a defined time t_1 , v at the entrance of the upper half section is set to $v \approx 1$ m/s. At a time $t_2 \approx t_1 + 0.5$ s, fluid velocity in the lower half section is also set to $v \approx 1$ m/s. Experimentally, this time shift is realised by a blocking mechanism, which covers the lower half for 0.5 s. By monitoring the front surface of the sample with an infrared camera, the temperature distribution during the cooling phase can be recorded.

2.4. Thermal conductivity

Thermal conductivity measurements have been performed with the so-called *transient heat source technique* [8], also well known as “hot disk” method, which uses a thin nickel double

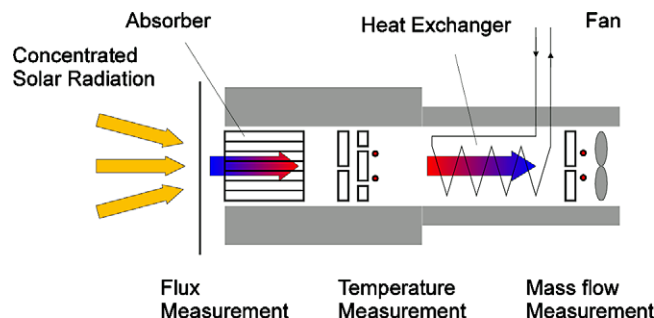


Fig. 1. Test bed used for open volumetric receiver efficiency measurement.

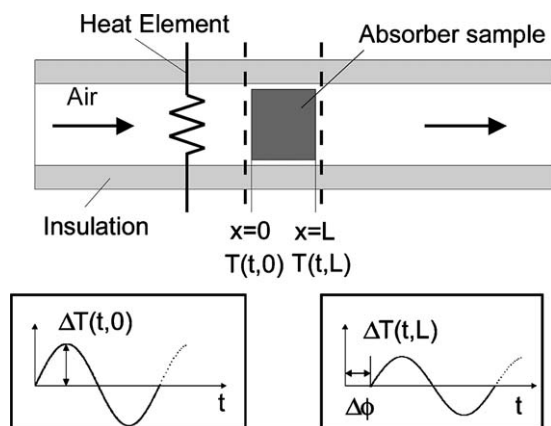


Fig. 2. Experimental set-up for measuring convective heat transfer properties of porous materials.

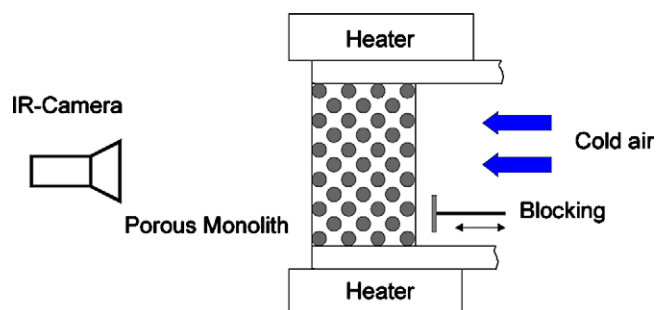


Fig. 3. Experimental set-up for investigating the front temperature distribution of a heated porous monolith cooled with ambient air.

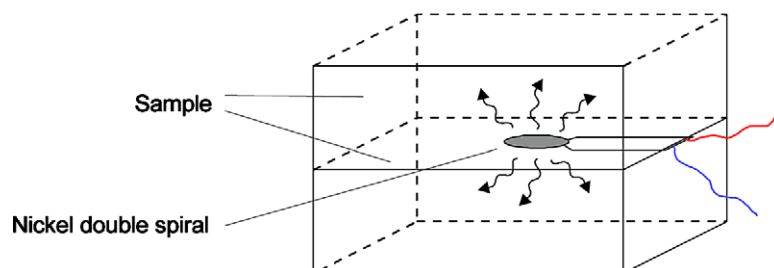


Fig. 4. *Transient heat source technique* to measure the thermal conductivity of porous materials.

spiral as a heat source, placed between two identical material samples. Simultaneously, it is used as a sensor for temperature measurement, Fig. 4. Depending on pore size and sample dimensions, sensors of 10–50 mm are available. To avoid any influence of pore size level effects, the disk has to be much larger than the average pore diameter. While delivering a defined quantity of electrical power in a defined time period to the hot disk, the curve of the increasing temperature is recorded. The thermal conductivity of the sample material now influences the slope of the curve. A numerical fit of the temperature curve yields thermal conductivity Λ and thermal diffusivity κ of the sample material. Specific heat C is then given with the density ρ by $\Lambda/\rho\kappa$.

3. Materials investigated

3.1. IKTS foam ceramics

Under variation of thermal conductivity of the solid cell wall material and cell density (10, 20, 45 ppi (pores per inch)), a set of monolith samples has been manufactured by the Institut für Keramische Technologien und Sinterwerkstoffe in Dresden, Germany (Fig. 5). The variation of thermal conductivity has been realised by employing cordierite ($\lambda_{\text{WALL}} \approx 1 \text{ W/mK}$), siliconised silicon carbide (SSiC, $\lambda_{\text{WALL}} \approx 100 \text{ W/mK}$) and clay bound silicon carbide (CBSiC, $\lambda_{\text{WALL}} \approx 10 \text{ W/mK}$). How these values influence the thermal conductivity of the system wall/pore is indicated in Table 1.

3.2. SiC fibre mesh

SiC fibre mesh material originally was developed for burners. The material used in this test is manufactured by Schott Glas in Mainz, Germany and has the product name Ceramat® FN (Fig. 6). It consists of silicon carbide fibres of 25 μm diameter glued together to form a layer of 3.5 mm thickness. The fibres are oriented in directions perpendicular to the direction of the air flow, beneficial for radial heat transport and good heat transfer properties.

3.3. Ceramic (SiC) catalyst carriers

SiC monoliths manufactured by the Danish company HELIOTECH have been successfully applied in the European SOLAIR project in a 200 kW scale as a solar absorber. A channel



Fig. 5. Ceramic foam manufactured by the IKTS.

Table 1

Geometric and thermophysical properties of the investigated porous materials

Material	Absorptivity (%)	Specific surface (m^2/m^3)	Mean pore size/channel $\varnothing$ (mm)	Mean wall thickness/fibre $\varnothing$ (mm)	Heat conductivity (W/mK)
Fibre mesh	95.3	8000	–	0.025	0.08
SiC catalyst	96.0	1000	2	0.8	–
SSiC catalyst	90.4	1000	2	0.8	0.4
Metallic catalyst 800	–	4300	0.9	0.030	–
Metallic catalyst 1600	–	6100	0.6	0.020	–
SSiC foam 10 ppi	–	500	1.7	–	–
SSiC foam 20 ppi	–	1000	0.8	–	–
SSiC foam 45 ppi	–	2500	0.4	–	2.7
CBSiC foam 10 ppi	–	500	1.7	–	–
CBSiC foam 20 ppi	–	1000	0.8	–	–
CBSiC foam 45 ppi	–	2500	0.4	–	0.6
Cordierite foam 20 ppi	–	1000	0.4	–	0.2

width of 2 mm and a wall thickness of 0.8 mm lead to minor heat transfer in the front area of the material causing relatively high front temperatures. Therefore, the efficiency potential is limited. However, it has been applied because of its excellent resistance to high temperatures up to 1600 °C. Air outlet temperatures of over 1000 °C have been reached. In this study, it has been mainly used in combined absorber systems as a tool to achieve homogeneous flow.

3.4. Metallic catalyst carriers

Metallic catalyst carriers from the German company EMITEC are manufactured from a thin sheet high-temperature steel. Eight hundred and 1600 cpsi (cells per square inch) materials were available (Fig. 7). The material will be denoted as “Catrec” in the following. Additionally, an 800 cpsi material was modified by introducing small chokes in each channel to achieve a quadratic dependency of the pressure loss as a function of flow velocity.

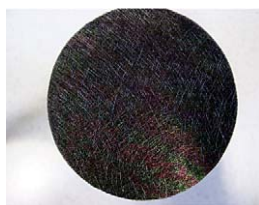


Fig. 6. Silicon carbide fibre mesh material Ceramat[®] FN manufactured by Schott Glas, Mainz, Germany.

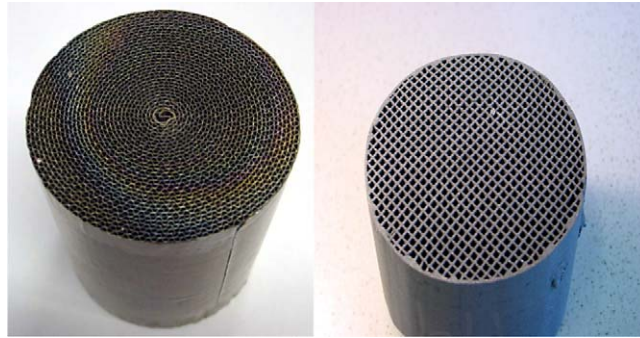


Fig. 7. Metallic (left) and silicon carbide ceramic catalyst carrier material.

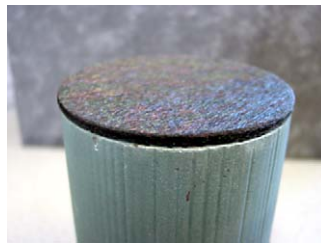


Fig. 8. Receiver element made out of an SiC fibre mesh and a SiC catalyst carrier.

3.5. Material combinations

A front material comprising a high specific surface, excellent absorption and high porosity to achieve a volumetric absorption of the concentrated solar radiation (Schott Ceramat[®] FN) has been combined with a material of beneficial thermal conductivity properties and a quadratic pressure loss characteristic (SiC catalyst carrier). For the tests in concentrated solar radiation, two materials have been glued into a ceramic tube without directly connecting them (Fig. 8).

4. Results and discussions

4.1. Efficiency

In the first experiment, as an improved volumetric receiver system, a SiC fibre mesh has been compared as a stand alone system to a combined version consisting of a mesh plus a SiC parallel channel monolith (see Section 3.4). The results are presented in Fig. 9. At air outlet temperatures of 300–550 °C, the system reaches efficiencies of 75–92%. This is mainly caused by excellent absorption properties and an enormous amount of specific surface. However, due to its short length of only 3.5 mm, radial heat transfer is suppressed, leading to an inhomogeneous

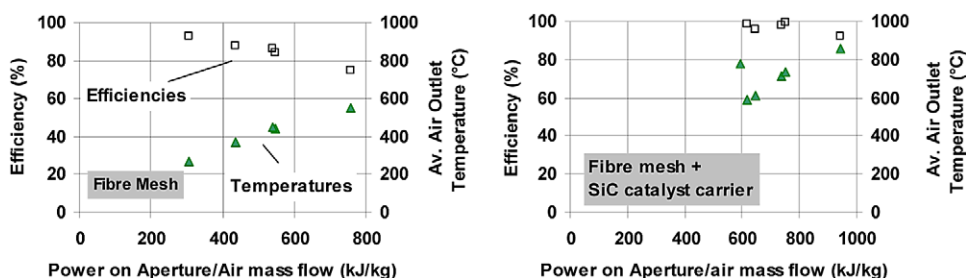


Fig. 9. Results of receiver efficiency measurements (fibre mesh, fibre mesh + SiC catalyst carrier).

temperature and flow distribution in the receiver sample as a consequence of the gaussian-like flux distribution of the concentrated solar radiation. In contrast, the combined system enables a radial heat transfer due to the good thermal conduction properties of the SiC monolith. Because the monolith also dominates the flow resistance properties of the system, a more homogeneous temperature distribution is reached, which generally leads to lower peak temperatures at the front part of the absorber material. Consequently, higher air outlet temperatures and higher efficiencies are reached.

Secondly, three metallic catalyst carriers have been compared, Fig. 10. Similar to the fibre mesh, flow predominantly penetrates through the colder parts of the receiver, a fact which is enhanced by the linear pressure drop characteristics of the material. To avoid peak temperatures over 1000°C , the average air outlet temperature had to be limited to 400°C with the 800 and 1600 cpsi material. Due to its higher specific surface, which leads to lower front temper-

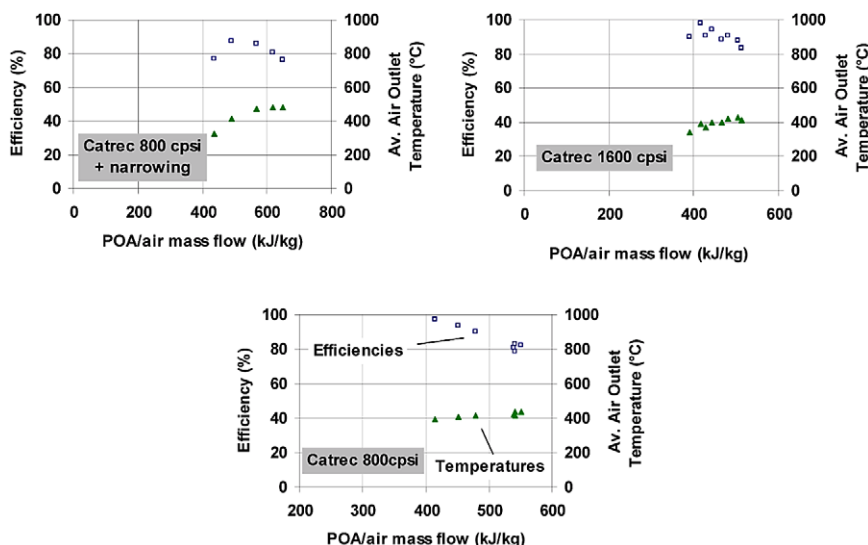


Fig. 10. Results of receiver efficiency measurements (metallic catalyst carriers).

atures, the 1600 cpsi material achieved 2–3% higher efficiencies. A slight improvement can be observed with the third carrier with the narrowing in between. The average air outlet temperatures achieved were a little higher due to a more homogeneous temperature distribution. However, as can be seen from pressure loss curves, the narrowing of the channels is not as distinctive as it is necessary for receiver application and further samples will be soon manufactured.

4.2. Heat transfer

Heat transfer properties of the foam materials (Section 3.1) are presented as the product of heat transfer coefficient α and specific surface $A_V\alpha$. Reynolds numbers were calculated from pore size air flow velocity v_{Pore} , viscosity η , density ρ and hydraulic diameter d_H by

$$Re = v_{\text{Pore}} \cdot \rho \cdot d_H / \eta.$$

The pore size level quantities s (pore diameter,) v_{Pore} and d_H are derived using the following equations [9] (PPI-numbers n_{PPI} usually are given by the foam manufacturer in sufficient accuracy)

$$s = 0.0254 / 1.5 \cdot n_{\text{PPI}}$$

$$v_{\text{Pore}} = v / P_0$$

$$d_H = \sqrt{(1 - P_0) \cdot \frac{4s^2}{3 \cdot \pi}}.$$

Fig. 11 summarises the results. As expected, an increasing heat transfer with increasing Reynolds number coefficient is observed for all materials. Due to their higher specific surface, the PPI 45 materials significantly show higher values of $A_V\alpha$. This is more pronounced for CBSiC, which is probably caused by closed pores observed in the SSiC.

4.3. Flow instability

As an example, the homogeneity of the temperature distribution during the cooling phase of a porous material, through which cold air is flowing, has been compared for two materials: a

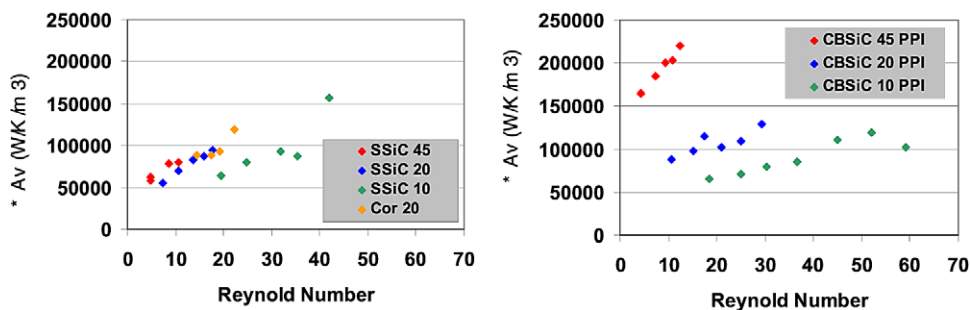


Fig. 11. Heat transfer properties of SSiC, CBSiC and cordierite ceramic foam materials.

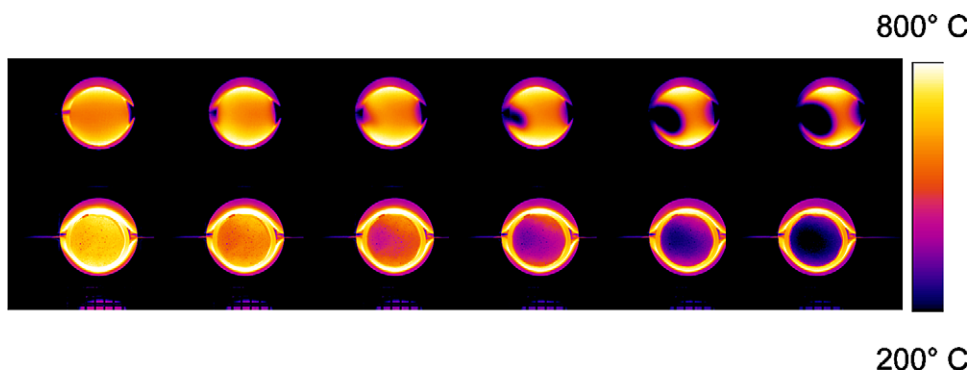


Fig. 12. Cooling behaviour of air-cooled pre-heated porous materials: a cordierite catalyst carrier (above) and a cordierite foam (below).

cordierite ceramic foam (20 ppi) and a cordierite catalyst carrier (400 cpsi) (Fig. 12). Though having similar bulk thermophysical properties, the architecture of the material leads to different pressure drop characteristics and thus to completely different flow properties. Slight temperature inhomogeneities observed at the front of the foam are rapidly compensated, whereas the catalyst carrier shows a permanent inhomogeneous temperature distribution.

5. Conclusions

As a basis for numerical investigations, thermal conductivity and heat transfer properties of selected materials have been experimentally determined. Additionally, the efficiency potential of receiver prototypes consisting of two materials has been shown in a laboratory scale. An experimental method of detecting unstable flow through porous media has been developed. This knowledge is presently exploited in two projects with the objective of considering the solar requirements during the manufacturing process. Most promising are materials based on ceramic foams and ceramic fabrics because of their potential towards large specific surface combined with beneficial pressure loss characteristics. The second important objective is to numerically predict the behaviour of these materials when applied in larger scale concentrated solar radiation. A possible introduction of these materials in solar tower technology will offer the potential for a more effective and reliable operation compared to the current open volumetric receiver technology.

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